



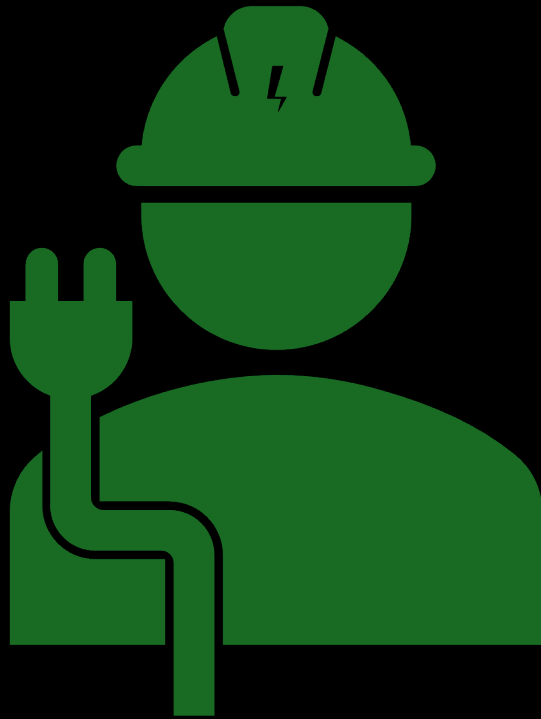
ISDT

Integrated Systems Design Team

SmartZone

Integrated Systems Design Team:

Sabrina, Kyle, Annika

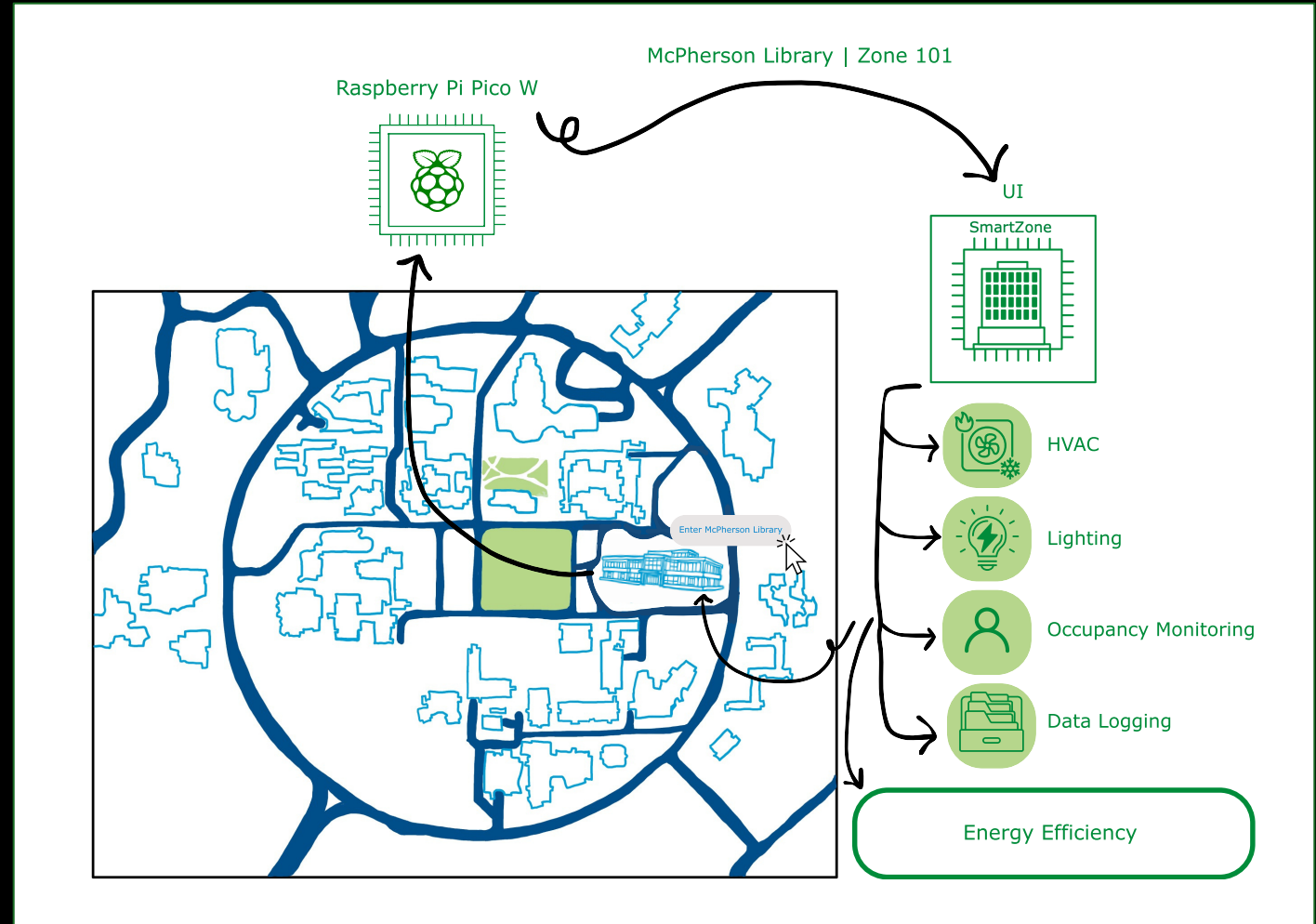


What Problem Are We Solving?

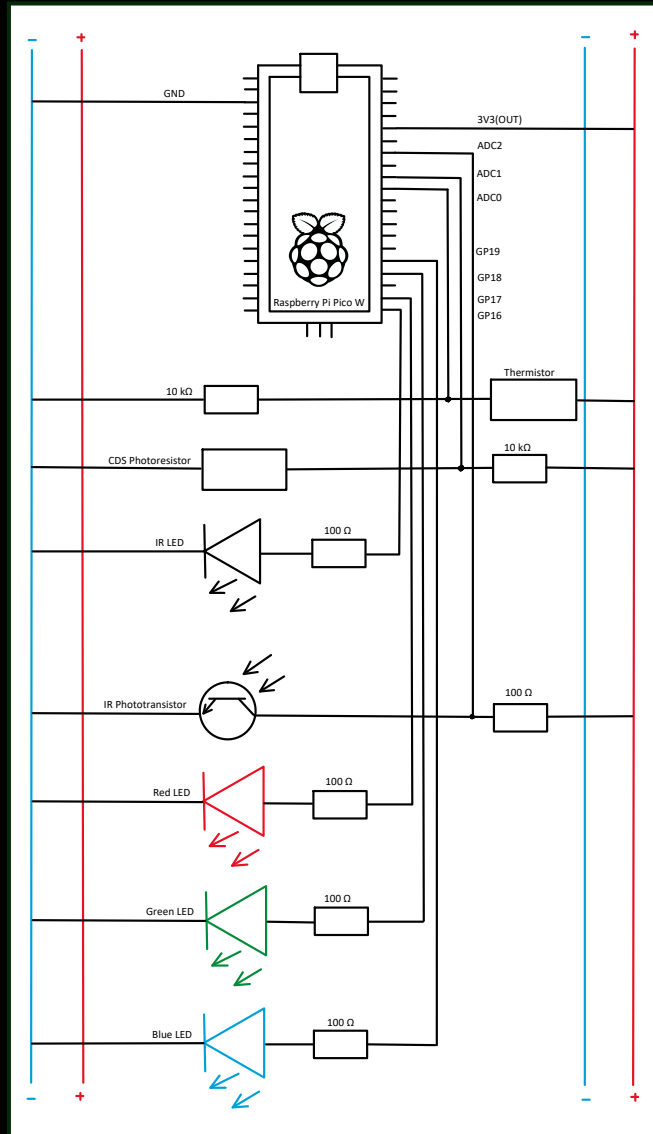
- **Heating, Cooling, and Lighting** Systems across UVic's campus operate in isolation, driven by schedules
- UVic's current gap in building management: fragmented control and limited environmental responsiveness
- RFP-202601 calls for **Integrated Building Management**
- Chosen for alignment with UVic's current green campus initiatives
- Research indicated that HVAC and lighting systems offer the best energy saving potential

Our Proposed Solution: SmartZone

- Built to address the current **fragmented building control** and improve **energy efficiency**.
- An integrated, **occupancy-aware heating, cooling, and lighting** system that responds to real-time building zone conditions.
- Monitors ambient light, temperature, and occupancy.
- Operates lighting (green), heating (red), and cooling (blue) LED indicators.
- Provides **user-friendly** monitoring and manual override through the UI.
- Includes a preliminary log function to demonstrate long-term energy optimization capabilities
- This solution was selected as it directly targets energy wasted in unoccupied spaces, aligning with **UVic's current green initiatives**.

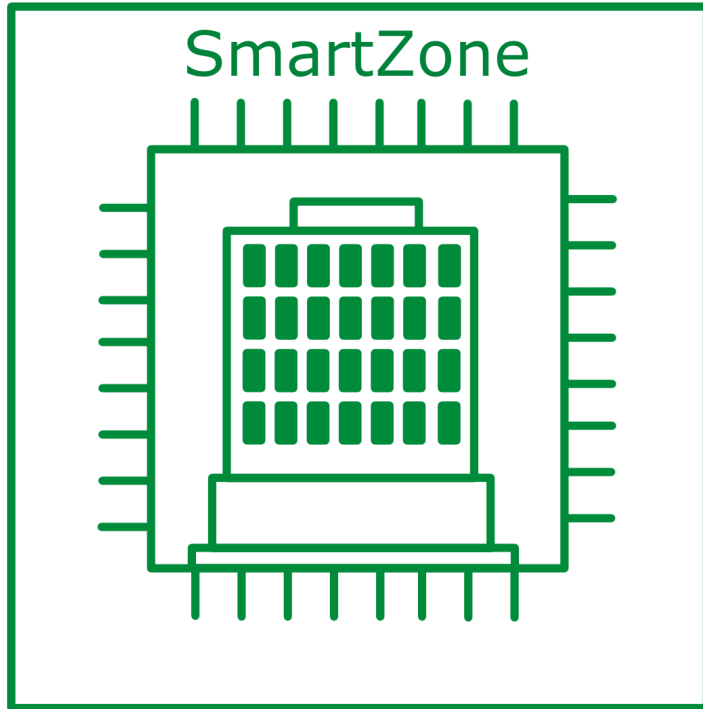


How is ISDT Solving the Problem?

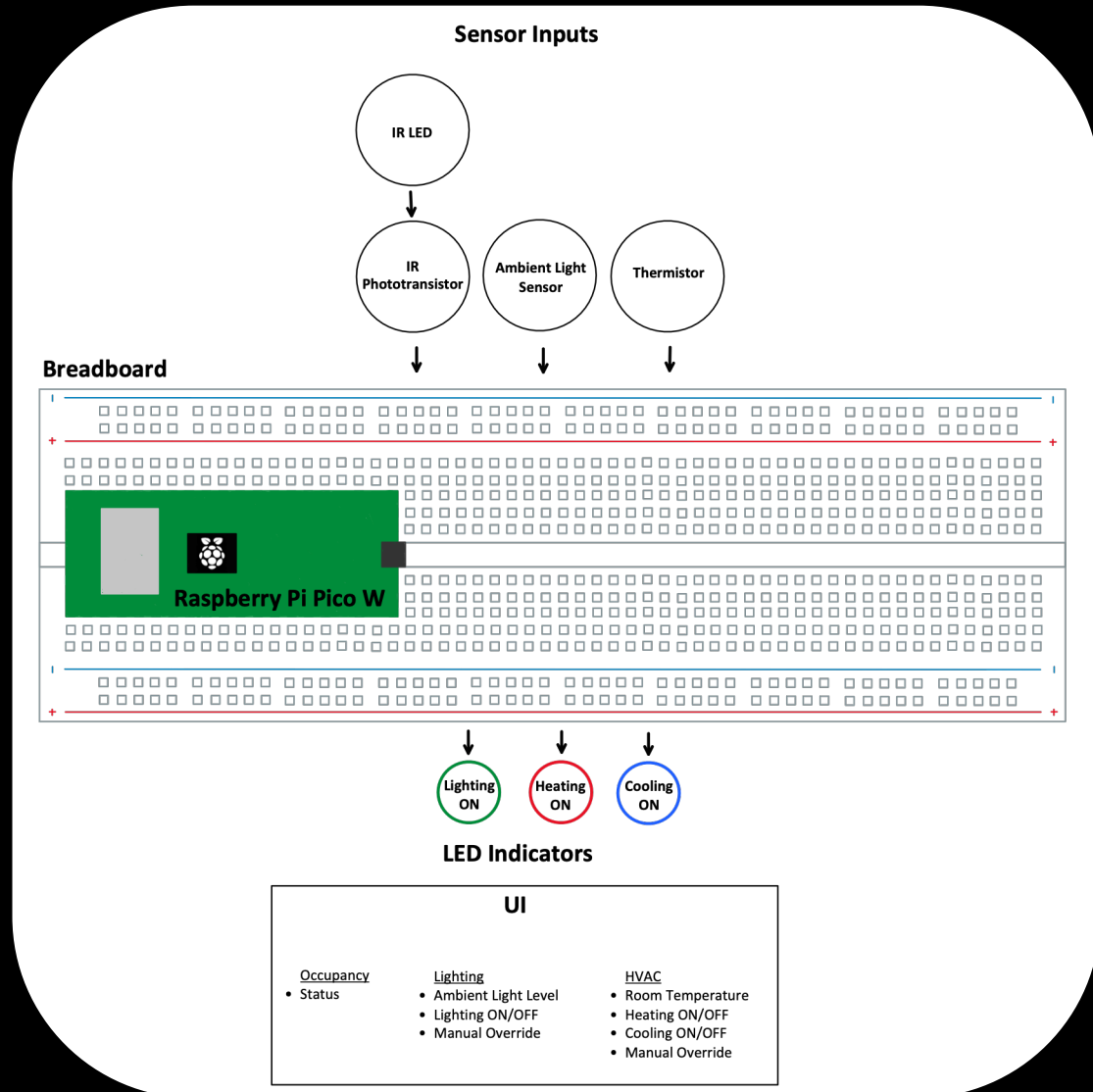


Component	Quantity	Need
Raspberry Pi Pico W (SC0915)	1	Central microcontroller which processes sensor input, hosts web UI, and executes actuator indicator activation.
NTC Thermistor (TH310J39G)	1	Measures ambient temperature for heating and cooling control using voltage divider.
CDS Photoresistor (MJ5516)	1	Measures ambient light level to enable daylight responsive lighting control.
IR Phototransistor (LTR-3208E)	1	Detects infrared beam interruption to determine occupancy state.
IR LED (IR333-A)	1	Emits infrared signal for occupancy detection.
Green LED (WP7113GD)	1	Indicates lighting actuator activation state.
Red LED (WP7113ID)	1	Indicates heating actuator activation state.
Blue LED (CHANZON 100F5W-YT-BL-BL)	1	Indicates cooling actuator activation state.

Hardware Logic Flowchart

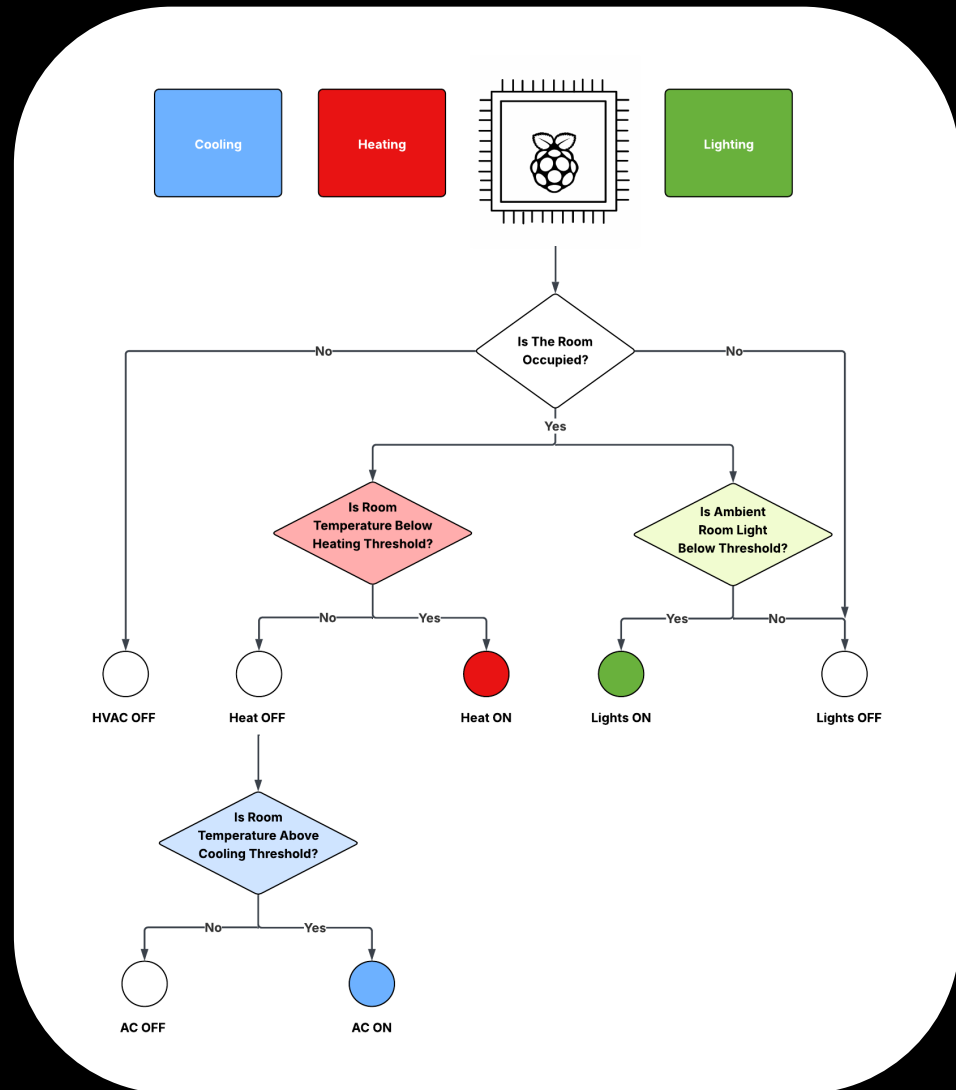


Simplified Systems Flowchart



- Occupancy Monitoring
- Ambient Light Sensing
- Temperature Sensing

Control Logic Flowchart



LED indicators activate according to zone conditions:

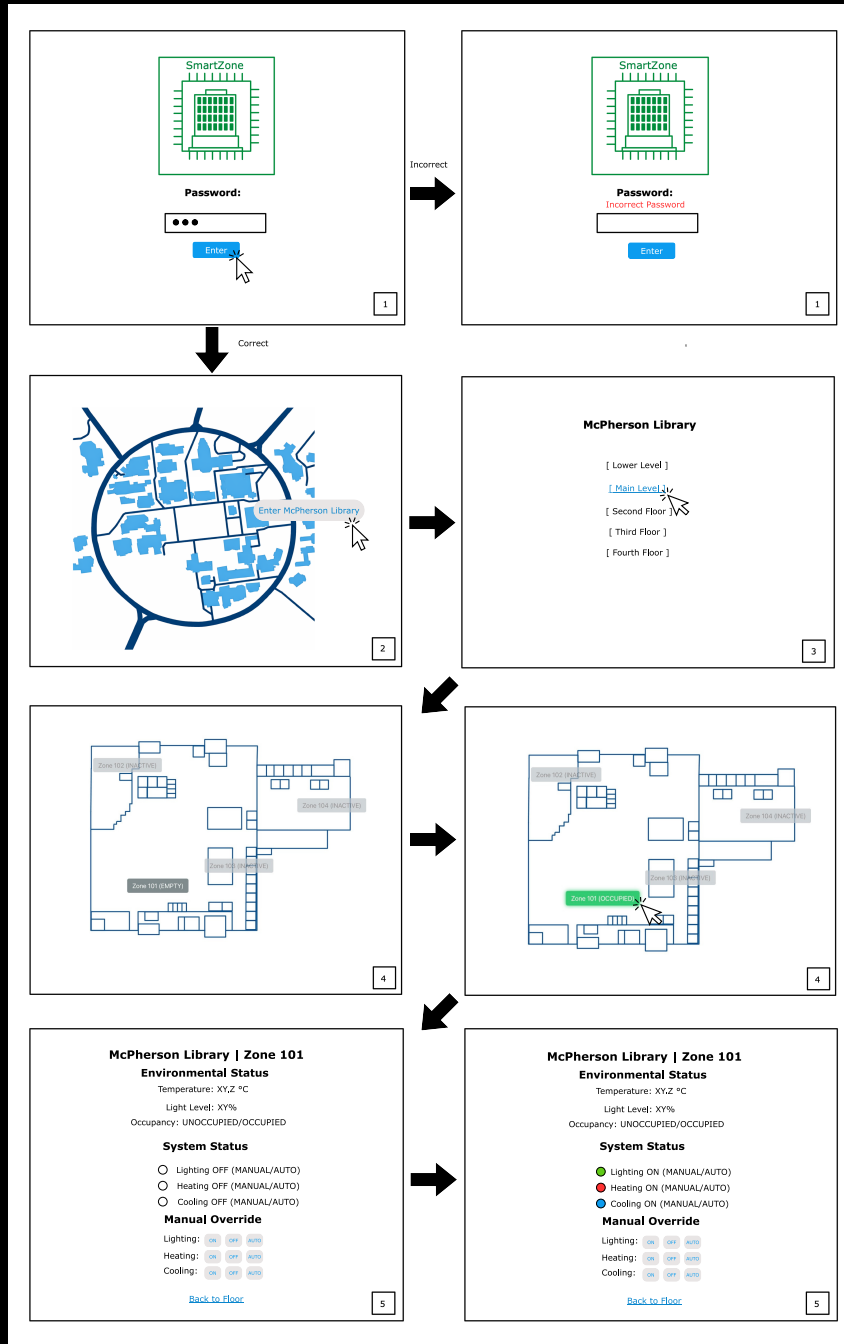
- Lighting (Green):
 - ON: Occupied & Daylight < Threshold
 - OFF: Unoccupied or Daylight > Threshold
- Heating (Red):
 - ON: Occupied & Temperature < Heating Threshold
 - OFF: Unoccupied or Temperature > Heating Threshold
- Cooling (Blue):
 - ON: Occupied & Temperature > Cooling Threshold
 - OFF: Unoccupied or Temperature < Cooling Threshold

SmartZone UI Flowchart

- Universal Design Principles

#	Principle	Method of Implementation	Satisfied
1	Equitable use	Uses dyslexia friendly font (Verdana) to avoid stigmatizing users.	Yes
2	Flexibility in Use	Provides user with the choice of passive monitoring (through system display) or active control (manual overrides).	Yes
3	Simple and Intuitive Use	Uses visual navigation (image of Ring Road and McPherson Library floorplan) to eliminate complexity.	Yes
4	Perceptible Information	Presents different modes of information via written and visual status indicators.	Yes
5	Tolerance for Error	Provides fail safe measures through manual override controls for each system.	Yes
6	Low Physical Effort	Enables entirely automated building control, eliminating the need for physical interaction.	Yes
7	Size and Space for Approach and Use	Ensures equitable use by not implementing any spatial aspects.	Yes
Final Accessibility Score			7/7

- Two-Way Communication: Manual Overrides (ON/OFF)



Design Considerations

1) **Reliability:** Positive testing response, test.py and config.py allow for easy testing and calibration

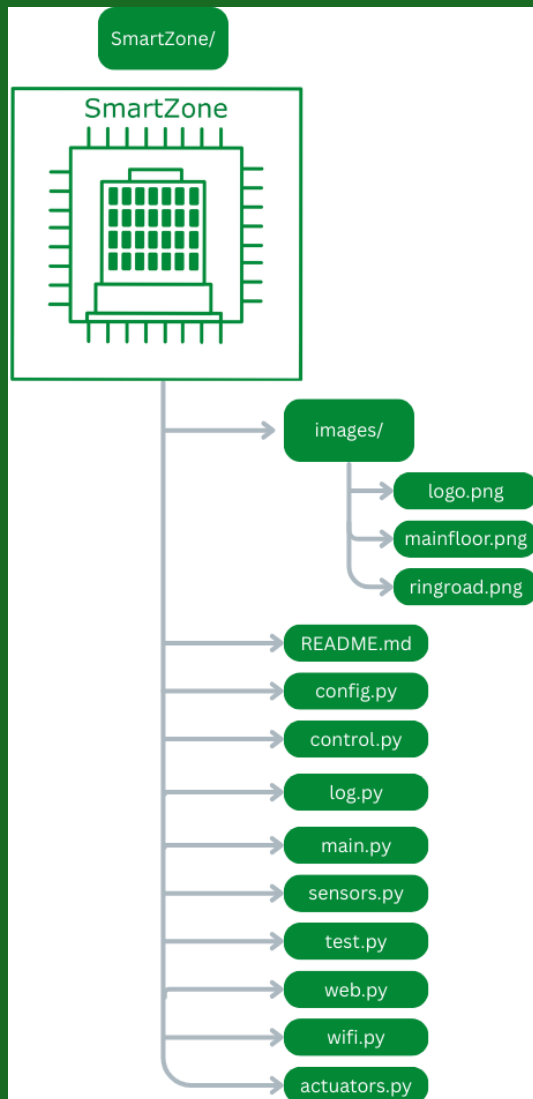
2) **Robustness:** Visualizes UVic to show scalability, structured file system allows for easy integration, multi-system integrated function

3) **Redundancy:** Dual password protection, swap potentiometer for other sensor in the case of failure (to simulate response)

4) **Safety:** Manual overrides, password protected

5) **User-friendliness:** Simple navigation, dyslexia friendly, modular repository with codes and documentation

Software Logic Flowchart



- **Images/**
 - logo.png: SmartZone logo
 - mainfloor.png: McPherson Library mainfloor layout
 - ringroad.png: Campus map used in system UI
- **README.md**: Provides setup instructions and system overview
- **config.py**: Stores global variables and system settings.
- **control.py**: Implements control algorithm to activate LED responses.
- **log.py**: Enables data logging by writing system variables to a csv file every 60 seconds.
- **main.py**: Runs the SmartZone system.
- **sensors.py**: Processes temperature, occupancy, and daylight inputs.
- **test.py**: Allows the user to see the environmental conditions and set the configurations accordingly.
- **wifi.py**: Handles network connection.
- **actuators.py**: Activates LEDs to indicate system activation when called.

i) ISDT SmartZone Progress

Category	06-Feb	13-Feb	06-Mar	13-Mar	20-Mar
Circuit Designing					
Sensor Coding					
UI Coding					
Integration					
Calibration + Documentation					

All deliverables 100% completed, additional features added

Design and testing results are documented within Team Recommendation Reports

Code with comments is documented in GitHub:
<https://github.com/sabrinabryce/SmartZone>

SmartZone runs full functionality

ii) Testing

1. Real-Time Visibility

Purpose: Evaluate whether the SmartZone UI updates within the programmed **1-second refresh interval**.

Procedure:

1. Launch the SmartZone dashboard.
2. Monitor UI refresh intervals while the system is running.
3. Record the refresh time for each trial.
4. Mark each trial as **Pass** if the refresh occurs within 1 second, or **Fail** if it exceeds the limit.
5. Repeat for **10 trials**.

Trial #	Update Time Taken (s)	Refresh Within Time? (Pass/Fail)
1	2.4	F
2	1.6	F
3	2.3	F
4	1.0	P
5	1.0	P
6	2.4	F
7	1.0	P
8	1.3	F
9	1.0	P
10	1.0	P
Pass Rate		50%

Observation:

The SmartZone UI occasionally exceeded the 1-second refresh target. This delay is likely due to the **Raspberry Pi Pico W operating near its memory capacity**, as the UI includes multiple graphical elements such as JPG images.

ii) Testing

2. Condition Response Accuracy

Purpose: Verify that the system activates **heating, cooling, and lighting** according to environmental conditions.

Procedure:

1. Simulate environmental inputs using sensors or controlled inputs.
2. Observe LED indicators representing system outputs.
3. Compare **expected system response** with the **actual system response**.
4. Record results as **Pass or Fail**.
5. Conduct **10 trials per system**.
6. Calculate average response accuracy.

Lighting System Accuracy

Trials	Pass	Fail	Accuracy
10	10	0	100%

Cooling System Accuracy

Trials	Pass	Fail	Accuracy
10	9	1	90%

Heating System Accuracy

Trials	Pass	Fail	Accuracy
10	9	1	90%

Observation:

One trial showed a **delayed cooling and heating activation**, resulting in a failure.

ii) Testing

- **3. Integrability**
- Purpose: Evaluate how many building systems operate correctly within the SmartZone platform.
- Procedure:
 - Evaluate heating, cooling, and lighting systems individually.
 - Determine system accuracy based on trial results.
 - Classify systems as **functional if accuracy exceeds 50%**.

System	% Accuracy	Functional? (>50%)
Lighting	100%	✓
Cooling	90%	✓
Heating	90%	✓
Functional Systems:	93.3%	3

ii) Component Verification Testing

4. Component Verification

Purpose: Confirm that hardware components and software features operate correctly.

Tests included:

- Sensor functionality verification
- UI navigation testing
- Manual override testing
- Failure condition testing

Deliverable	Complete?	Tested By	Test Method	Failure Testing?
Temperature Circuit	Yes	All	Multimeter voltage verification + LED trigger in Thonny	Yes: disconnected sensor to confirm error state
Ambient Light Circuit	Yes	All	Cover sensor + flashlight test	Yes: removed sensor to confirm % drops to zero
IR Occupancy	Yes	All	Object detection test + UI update	Yes: no object test + forced false input
UI Navigation	Yes	All	Cross-device browser test	Yes: wrong password test
Manual Override	Yes	All	Force LEDs off then override via UI	Yes: override during active sensor state